Risk-Aware Deep Reinforcement Learning for Daily Portfolio Allocation

B.Tech Project-I report submitted to

Indian Institute of Technology Kharagpur

in partial fulfilment for the award of the degree of

Bachelor of Technology (B. Tech)

in

Industrial and Systems Engineering

by

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Autumn Semester, 2026–27

November 2026

DECLARATION

I certify that

(a) The work contained in this report has been done by me under the guidance of my supervisor.

(b) The work has not been submitted to any other Institute for any degree or diploma.

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CERTIFICATE

This is to certify that the project report entitled "**Risk-Aware Deep Reinforcement Learning for Daily Portfolio Allocation**" submitted by **Ritam Mondal** (Roll No. 23IM30018) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Bachelor of Technology in Industrial and Systems Engineering is a record of bona fide work carried out under my supervision and guidance during Autumn Semester, 2026–27.

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Acknowledgement and Abstract

Acknowledgement:

I express my sincere gratitude to my supervisor, Dr. Abhishek Sharma, for his insightful guidance, intellectual mentorship, and continuous encouragement throughout this B.Tech. Project. His domain expertise in financial engineering, operations research, and sequential decision processes provided invaluable direction for this study. I also thank the Department of Industrial and Systems Engineering, Indian Institute of Technology Kharagpur, for providing computational facilities, and my peers and family for their continuous support.

Abstract:

Automating daily portfolio allocation via deep reinforcement learning (DRL) requires balancing return generation against market non-stationarity, partial state observability, and execution frictions. Prior literature separately suggests temporal encoders, risk-sensitive rewards, and turnover penalties, but these mechanisms are rarely evaluated under a single controlled setup. This report investigates a four-stage ablation: (i) Model M1: PPO with a feedforward MLP baseline and net return reward; (ii) Model M2: PPO with a recurrent LSTM encoder; (iii) Model M3: LSTM-PPO with recursive Differential Sharpe Ratio (DSR) reward; and (iv) Model M4: LSTM-PPO with DSR reward and a quadratic weight-change penalty ($\lambda_{\text{turn}} = 0.1$). All agents output continuous Dirichlet allocations over ten liquid National Stock Exchange of India (NSE) equities and an assumed zero-return cash reserve.

Models are evaluated across four chronological walk-forward folds spanning 2018–2025 across three random seeds under realistic proportional transaction costs (0.1%). In our experimental setup, the LSTM encoder (M2) improves upon the MLP baseline (M1) in pooled Sharpe ratio (0.92 vs. 0.71) and mean two-year test return (28.2% vs. 20.6%). The DSR reward (M3) demonstrates mixed, regime-dependent effects without an overall advantage over M2. Turnover regularisation (M4) effectively curtails daily turnover (cutting costs from 4.14% to 0.97% of capital) and achieves the shallowest maximum drawdown (−16.04%), but trades off return. Crucially, the passive Equal-Weight benchmark outperforms all RL models in mean two-year test return (36.48%) and Sharpe ratio (1.13), driven by continuous broad-market exposure in a secular bull market and cash drag from the agents' learned liquidity holdings.

Keywords: Deep Reinforcement Learning, Portfolio Allocation, Proximal Policy Optimisation, Dirichlet Distribution, Differential Sharpe Ratio, Turnover Regularisation, Walk-Forward Validation.

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Chapter 1 — Introduction

1.1 Background and Motivation

Daily portfolio allocation is a sequential decision-making problem in which asset weights must be continuously adjusted as market prices and conditions change. An effective strategy must balance return generation, financial risk, and transaction costs. However, relying only on current-day market indicators may overlook important temporal patterns, while maximising raw returns may not adequately account for risk. Frequent portfolio rebalancing can also increase transaction costs and reduce net returns.

This study investigates three complementary mechanisms within a common Proximal Policy Optimisation (PPO) framework: temporal memory through a Long Short-Term Memory (LSTM) encoder, risk-aware reward shaping using the Differential Sharpe Ratio (DSR), and turnover control through a quadratic weight-change penalty. Their individual contributions are examined through four controlled models (M1–M4) using chronological walk-forward evaluation on ten NSE equities, with a passive Equal-Weight portfolio as the benchmark.

Chapter 2 — Literature Review

Deep reinforcement learning for portfolio management draws on advances in policy optimisation, temporal modelling, risk-sensitive rewards, and continuous portfolio allocation. The following studies directly motivate the methodology adopted in this work.

Schulman et al. (2017) [4] introduced Proximal Policy Optimisation (PPO), using a clipped surrogate objective to restrict excessively large policy updates. This provides the common policy optimisation framework used across all four models in this study.

Zou et al. (2024) [7] proposed an equity trading architecture that combines a recurrent LSTM encoder with PPO policy optimization. Using a 30-day historical window and a 512-dimensional hidden representation, their system showed that multi-day sequential features capture price momentum more effectively than single-day snapshots. However, their policy rewards focused on nominal trading profit for individual equities rather than risk-adjusted continuous portfolio allocation across a multi-asset universe.

Millea (2021) [2] reviewed reinforcement learning approaches in quantitative trading, including the Differential Sharpe Ratio (DSR) for online risk-adjusted reward estimation. This study adopts the recursive DSR formulation to investigate risk-aware reward shaping in Models M3 and M4.

Liu et al. (2024) [1] introduced FinRL-Meta, providing a framework for financial reinforcement learning and chronological walk-forward evaluation. Its rolling train–validation–test approach motivates the four-fold chronological evaluation protocol used in this study.

Wang and Liu (2025) [5] investigated risk-sensitive reinforcement learning for portfolio allocation, incorporating downside-risk and turnover controls. Their work motivates the investigation of explicit turnover regularisation through the quadratic weight-change penalty in Model M4.

Yang et al. (2022) [6] introduced a continuous portfolio allocation approach based on the multivariate Dirichlet distribution. This study adopts the Dirichlet policy to generate non-negative portfolio weights that naturally satisfy the unit-simplex constraint across ten equities and a cash allocation.

Chapter 3 — Research Gap and Problem Formulation

3.1 Identified Research Gap

While significant progress has been made across temporal modeling, risk-aware reward design, and evaluation standards, existing studies in financial reinforcement learning typically evaluate these mechanisms in isolation. In the reviewed literature, different papers employ distinct equity universes, divergent transaction-cost assumptions, and incompatible backtesting horizons. Moreover, when advanced models are proposed, multiple architectural changes, loss modifications, and execution constraints are frequently introduced simultaneously.

Consequently, it remains difficult to determine whether reported performance improvements arise from the recurrent sequence encoder, the risk-adjusted reward structure, the turnover regulariser, or specific dataset artifacts. Table 3.1 positions this study relative to the reviewed literature, highlighting the single-factor controlled ablation methodology adopted here.

Table 3.1: Positioning of this work relative to the reviewed literature.

Paper	Temporal Memory	Risk-Aware / Reward	Turnover Control	Continuous Allocation	Controlled Ablation
Schulman et al. (2017) [4]				✓ *(PPO benchmark)*	
Millea (2021) [2]		✓ *(DSR survey)*			
Zou et al. (2024) [7]	✓				
Liu et al. (2024) [1]				✓ *(portfolio weights)*	
Wang & Liu (2025) [5]		✓ *(risk-sensitive)*	✓	✓	
Yang et al. (2022) [6]				✓ *(Dirichlet policy)*	
This work	✓	✓ *(DSR)*	✓ *(turnover penalty)*	✓	✓

3.2 Research Objectives

To address this gap through controlled empirical evaluation, the specific objectives of this project are:

1. **Controlled Four-Stage Ablation:** To implement a single-factor ablation framework consisting of four models (M1: MLP + net return; M2: LSTM + net return; M3: LSTM + DSR; M4: LSTM + DSR + quadratic penalty) to isolate the separate contributions of temporal memory, risk-aware reward shaping, and turnover regularisation.
2. **POMDP Trading Formulation:** To formulate daily portfolio allocation as a Partially Observable Markov Decision Process (POMDP), incorporating causal technical indicators, continuous Dirichlet policy sampling on the simplex manifold, and realistic transaction-cost deduction.
3. **Rigorous Chronological Evaluation:** To evaluate all agents across an eight-year walk-forward test protocol (2018–2025) spanning four folds and three random seeds, comparing performance against a passive Equal-Weight benchmark.

3.3 Research Questions

This investigation focuses on answering one central research question:

> *"How do temporal memory, risk-aware reward shaping, and turnover regularization independently and cumulatively affect the risk-adjusted out-of-sample performance of a PPO-based daily trading agent?"*

We investigate four specific sub-questions:

- **RQ1 (Temporal Architecture):** How does an LSTM recurrent encoder alter out-of-sample risk-adjusted return and daily turnover compared with a static MLP baseline?
- **RQ2 (Risk-Aware Reward Shaping):** What is the empirical effect of replacing net return with the recursive online Differential Sharpe Ratio (DSR)?
- **RQ3 (Turnover Regularisation):** Does augmenting the reward with a quadratic weight-change penalty reduce turnover and transaction costs, and what is the trade-off with return?
- **RQ4 (Benchmark Competitiveness):** How do the learned active RL allocation policies compare against a passive, fully-invested Equal-Weight benchmark under identical transaction frictions?

3.4 Stock Market Environment and POMDP Formulation

We model the trading process as an interaction between an agent and a simulated stock-market environment. At the close of each trading day t, the agent observes historical market information and current asset holdings, decides on target portfolio weights, and executes trades. The environment deducts proportional transaction costs, updates portfolio wealth based on subsequent price changes, and emits a scalar reward. This environment operates as an offline historical simulation rather than a live execution system.

Because daily market observations provide an incomplete reflection of the broader economic system, we formalise the allocation task as a Partially Observable Markov Decision Process (POMDP) [2], defined by the 7-tuple:

$$\mathcal{P} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \Omega, \mathcal{O}, \gamma \rangle \quad [2], \tag{3.1}$$

where $\mathcal{S}$ is the unobservable true market state, $\mathcal{A}$ is the action space, $\mathcal{T}$ represents transition dynamics, $\mathcal{R}$ is the reward function, Ω is the observation space, $\mathcal{O}$ is the observation probability distribution, and $\gamma \in [0, 1)$ is the discount factor.

In real financial markets, the true state $\mathcal{S}$ includes unobservable variables such as order-book queues, private liquidity flows, macroeconomic expectations, and trader sentiment. The agent only receives an observation vector $\mathbf{s}_t \in \Omega$ drawn from $\mathcal{O}(\mathbf{s}_t \mid \mathcal{S}_t)$, consisting of public OHLCV data and technical indicators. Because single-day observations do not satisfy the Markov property, recurrent neural networks (LSTM) are employed to summarize historical observation sequences $\mathbf{F}_t = [\mathbf{s}_{t-T+1}, \ldots, \mathbf{s}_t]$ spanning time steps $t - T + 1$ to t into a learned internal representation $\mathbf{h}_t$, directly mitigating partial observability [7].

3.4.1 State Space

The instantaneous observed market and portfolio state at trading day t is denoted by:

$$\mathbf{s}_t \in \mathbb{R}^{141}. \tag{3.2}$$

Its 141 dimensions are composed of:

- 13 technical features computed for each of the $N = 10$ equities ($13 \times 10 = 130$ market features).
- 11 current portfolio weights $\mathbf{w}_t^{\text{cur}} \in \mathbb{R}^{11}$, representing the allocations held in the 10 risky equities and 1 uninvested cash account at the start of day t.

Thus, $\mathbf{s}_t = [\mathbf{x}_t^\top, (\mathbf{w}_t^{\text{cur}})^\top]^\top$, where $\mathbf{x}_t \in \mathbb{R}^{130}$ contains market indicators and $\mathbf{w}_t^{\text{cur}} \in \mathbb{R}^{11}$ captures the portfolio state.

3.4.2 Action Space

The action vector $\mathbf{a}_t$ represents target portfolio allocations across the 10 risky equities and 1 cash asset:

$$\mathbf{a}_t = [a_{1,t}, a_{2,t}, \ldots, a_{10,t}, a_{\text{cash},t}]^\top \in \Delta^{10} \quad [6], \tag{3.3}$$

where Δ^{10} denotes the 11-dimensional unit simplex manifold [6]:

$$\Delta^{10} = \left\{ \mathbf{a}_t \in \mathbb{R}^{11} \;\middle|\; a_{i,t} \geq 0 \; \forall i \in \{1, \ldots, 11\}, \quad \sum_{i=1}^{11} a_{i,t} = 1 \right\} \quad [6]. \tag{3.4}$$

Every target weight is non-negative, all weights sum to one, short selling is prohibited, and cash is treated as an explicit allocation leg that allows the agent to de-risk. Following Yang et al. [6], this constraint is satisfied by parameterising the policy over a continuous multivariate Dirichlet distribution.

3.4.3 Reward Design

The scalar reward r_t emitted by the environment guides policy learning and defines our ablation dimensions:

- **Models M1 and M2 (Net Return):** Reward equals the realised net portfolio rate of return after deducting transaction fees: $r_t = R_{\text{net},t}$.
- **Model M3 (Differential Sharpe Ratio):** Reward equals the recursive online Differential Sharpe Ratio: $r_t = D_t$, penalising return variance [2].
- **Model M4 (DSR + Quadratic Penalty):** Reward equals the DSR reward minus an explicit quadratic weight-change regulariser on risky assets: $r_t = D_t - \lambda_{\text{turn}} \sum_{i=1}^{10} (w_{i,t}^{\text{target}} - w_{i,t}^{\text{cur}})^2$ [5].

The detailed mathematical formulations of these rewards are presented in Section 5.4.

Chapter 4 — Dataset and Experimental Methodology

4.1 Equity Universe and Data Acquisition

The experimental asset universe consists of $N = 10$ major large-cap equities actively traded on the National Stock Exchange of India (NSE), spanning diverse economic sectors: Reliance Industries (`RELIANCE.NS`, Energy/Conglomerate), Tata Consultancy Services (`TCS.NS`, IT Services), HDFC Bank (`HDFCBANK.NS`, Commercial

Banking), ICICI Bank (`ICICIBANK.NS`, Commercial Banking), ITC Limited (`ITC.NS`, Diversified FMCG), Hindustan Unilever (`HINDUNILVR.NS`, FMCG), Larsen & Toubro (`LT.NS`, Engineering & Infrastructure), Sun Pharmaceutical (`SUNPHARMA.NS`, Healthcare), Maruti Suzuki (`MARUTI.NS`, Automotive), and Bharti Airtel (`BHARTIARTL.NS`, Telecommunications). In addition, an uninvested cash reserve (Asset 11) is included to allow agents to de-risk.

Daily trading records were acquired from Yahoo Finance via `yfinance` spanning 1 January 2009 to 31 December 2025. Data acquisition used `auto_adjust=True`: Open, High, Low, and Close price series are adjusted backwards for stock splits and cash dividends, while volume is adjusted for splits. Historical data from 2009 to 2010 provides an initial burn-in period to initialise 200-day moving averages, ensuring that all technical features are fully stabilized by 1 January 2011. Across the common trading calendar from 2011 through 2025, there are 3,699 active trading days.

4.2 Feature Engineering and State Representation

For each asset $i \in \{1, \ldots, 10\}$ on trading day t, a 13-dimensional feature vector $\mathbf{f}_{i,t} \in \mathbb{R}^{13}$ is computed using causal transformations normalized by prevailing prices or rolling baselines, as summarized in Table 4.1.

Table 4.1: The 13 technical features computed daily for each equity asset ($i \in \{1, \ldots, 10\}$).

Index	Feature Name	Identifier	Mathematical Formulation
$f_{1,i,t}$	Daily Log Return	$r^{\log}_{i,t}$	$\ln(C_{i,t}/C_{i,t-1})$
$f_{2,i,t}$	Close-to-Open Relative Return	$o^{\mathrm{rel}}_{i,t}$	$C_{i,t}/O_{i,t} - 1$
$f_{3,i,t}$	High-to-Low Relative Spread	$hl^{\mathrm{rel}}_{i,t}$	$H_{i,t}/L_{i,t} - 1$
$f_{4,i,t}$	Normalized Relative Volume	$v^{\mathrm{rel}}_{i,t}$	$\ln(V_{i,t}/\mathrm{SMA}_{20}(V_{i,t}))$
$f_{5,i,t}$	50-day EMA Relative Deviation	$ema^{\mathrm{rel}}_{50,i,t}$	$C_{i,t}/\mathrm{EMA}_{50}(C_{i,t}) - 1$, smoothing $\alpha = 2/(50+1)$ [5]
$f_{6,i,t}$	200-day EMA Relative Deviation	$ema^{\mathrm{rel}}_{200,i,t}$	$C_{i,t}/\mathrm{EMA}_{200}(C_{i,t}) - 1$, smoothing $\alpha = 2/(200+1)$ [5]
$f_{7,i,t}$	Relative Strength Index	$\mathrm{RSI}^{\mathrm{norm}}_{i,t}$	$\mathrm{RSI}_{14,i,t}/100 - 0.5$, where $\mathrm{RS} = \bar{U}_{14}/(\bar{D}_{14} + 10^{-8})$ [5]
$f_{8,i,t}$	MACD Difference	$\mathrm{MACD}_{i,t}$	$(\mathrm{EMA}_{12}(C_{i,t}) - \mathrm{EMA}_{26}(C_{i,t}))/C_{i,t}$ [5]
$f_{9,i,t}$	MACD Signal Line	$\mathrm{Signal}_{i,t}$	$\mathrm{EMA}_9(\mathrm{MACD}^{\mathrm{raw}}_{i,t})/C_{i,t}$, where $\mathrm{MACD}^{\mathrm{raw}}_{i,t} = \mathrm{EMA}_{12}(C_{i,t}) - \mathrm{EMA}_{26}(C_{i,t})$ [5]
$f_{10,i,t}$	Average True Range	$\mathrm{ATR}^{\mathrm{rel}}_{i,t}$	$\mathrm{SMA}_{14}(\mathrm{TR}_{i,t})/C_{i,t}$, $\mathrm{TR}_t = \max(H_t - L_t, \lvert H_t - C_{t-1}\rvert, \lvert L_t - C_{t-1}\rvert)$ [5]
$f_{11,i,t}$	Bollinger Band %B	$\%B_{i,t}$	$(C_{i,t} - \mathrm{LB}_{i,t})/(\mathrm{UB}_{i,t} - \mathrm{LB}_{i,t} + 10^{-8})$, bands at $\pm 2\sigma_{20}$ [5]
$f_{12,i,t}$	Bollinger Bandwidth	$\mathrm{BW}_{i,t}$	$(\mathrm{UB}_{i,t} - \mathrm{LB}_{i,t})/\mathrm{MB}_{i,t}$ [5]
$f_{13,i,t}$	Average Directional Index	$\mathrm{ADX}^{\mathrm{norm}}_{i,t}$	$\mathrm{ADX}_{14,i,t}/100$

The instantaneous state vector $\mathbf{s}_t$ (or market observation vector $\mathbf{x}_t$) concatenates the 13 engineered technical features across all 10 equity assets ($10 \times 13 = 130$ market features) followed by the 11 current portfolio allocation weights ($w^{\mathrm{cur}}_{1,t}, \ldots, w^{\mathrm{cur}}_{10,t}$ for the equities and $w^{\mathrm{cur}}_{\mathrm{cash},t}$ for cash):

$$\mathbf{s}_t = \Big[\underbrace{f_{1,1,t}, \ldots, f_{13,1,t}}_{\text{Stock 1}}, \underbrace{f_{1,2,t}, \ldots, f_{13,2,t}}_{\text{Stock 2}}, \ \ldots, \ \underbrace{f_{1,10,t}, \ldots, f_{13,10,t}}_{\text{Stock 10}}, \ w^{\mathrm{cur}}_{1,t}, \ldots, w^{\mathrm{cur}}_{10,t}, w^{\mathrm{cur}}_{\mathrm{cash},t} \Big]^\top, \tag{4.1}$$

where $f_{j,i,t}$ denotes technical market feature j ($j \in \{1, \ldots, 13\}$) of equity asset i ($i \in \{1, \ldots, 10\}$) at time step t, $w^{\mathrm{cur}}_{i,t}$ represents the current allocation weight held in equity asset i, and $w^{\mathrm{cur}}_{\mathrm{cash},t}$ is the current uninvested cash reserve weight. Thus, the complete instantaneous state dimension is:

$$\boxed{\mathbf{s}_t \in \mathbb{R}^{141}} \qquad (10 \times 13 + 11 = 141). \tag{4.2}$$

For models utilizing temporal memory (M2–M4), the agent receives a sliding historical window spanning time steps $t - T + 1$ to t ($T = 30$ trading days) [7]:

$$\mathbf{F}_t = [\mathbf{s}_{t-T+1}, \mathbf{s}_{t-T+2}, \ldots, \mathbf{s}_t]^\top \in \mathbb{R}^{T \times 141} = \mathbb{R}^{30 \times 141} \quad [7]. \tag{4.3}$$

Feature standardization is fitted strictly on the training partition of each fold using mean $\boldsymbol{\mu}_{\mathrm{train}}$ and standard deviation $\boldsymbol{\sigma}_{\mathrm{train}}$, with standardized values clipped to $[-5.0, 5.0]$. Test partitions use training statistics without leakage.

4.3 Walk-Forward Evaluation Protocol

Following Liu et al. [1], the agents are evaluated using a 9-year rolling-origin walk-forward protocol for each of the four folds (spanning a contiguous 8-year out-of-sample test horizon from 2018 to 2025). Each fold consists of 6 years of training data, 1 year of validation data, and 2 years of held-out out-of-sample test data ($6 + 1 + 2 = 9$ years per fold), advancing chronologically by 2 years per fold (Table 4.2 and Figure 4.1).

Table 4.2: Chronological specification of the four walk-forward folds.

Fold	Training Period (6y)	Validation Period (1y)	Test Period (2y)	Train Days	Test Days
1	2011-01-01 to 2016-12-31	2017-01-01 to 2017-12-31	2018-01-01 to 2019-12-31	1,478	489
2	2013-01-01 to 2018-12-31	2019-01-01 to 2019-12-31	2020-01-01 to 2021-12-31	1,475	496
3	2015-01-01 to 2020-12-31	2021-01-01 to 2021-12-31	2022-01-01 to 2023-12-31	1,476	495
4	2017-01-01 to 2022-12-31	2023-01-01 to 2023-12-31	2024-01-01 to 2025-12-31	1,479	494

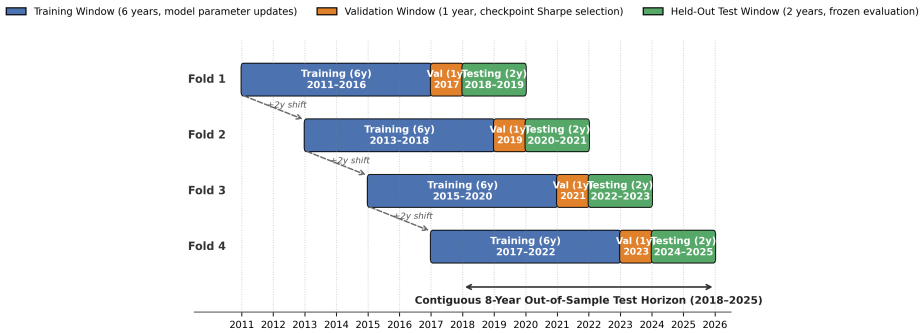


Figure 4.1: Chronological four-fold rolling walk-forward evaluation protocol with 6-year training, 1-year validation, and 2-year testing windows, advancing by 2 years per fold [1].

The test windows are mutually disjoint and together tile 2018 through 2025. It is important to distinguish the number of historical trading days in each split ($\approx 1,478$ training days and ≈ 490 test days) from the reinforcement learning training budget, which is fixed at 300,000 environment interaction steps across all models. Each configuration is trained across three fixed random seeds ($S \in \{0, 1, 2\}$). During training, validation Sharpe ratio is computed every 25,000 steps; the checkpoint achieving the highest validation Sharpe ratio is saved and evaluated once on the unseen test window.

4.4 Evaluation Measures

We evaluate out-of-sample portfolio performance using seven standard financial and operational metrics, summarized in Table 4.3.

Table 4.3: Summary of evaluation measures used in out-of-sample portfolio assessment.

Metric	Mathematical Definition	Interpretation / Citation
Cumulative Return (CR)	$CR = \frac{V_N}{V_0} - 1 = \prod_{t=1}^{N}(1 + R_{\text{net},t}) - 1$	Total percentage growth in portfolio wealth [7].
CAGR	$CAGR = (1 + CR)^{252/N} - 1$	Annualised geometric rate of return (252 trading days/year).
Sharpe Ratio (SR)	$SR = \frac{\bar{r} - r_f}{s_r}\sqrt{252}, \quad r_f = 0$	Risk-adjusted excess return per unit of volatility [2].
Sortino Ratio	$\text{Sortino} = \frac{\bar{r} - r_f}{\text{downside}}\sqrt{252}, \quad \text{downside} = \sqrt{\frac{1}{N}\sum_{t=1}^{N}\min(R_{\text{net},t}, 0)^2}$	Downside risk-adjusted return penalising negative deviations.
Maximum Drawdown	$MDD = \min_{t \in \{1,\ldots,N\}}\left(\frac{V_t}{\max_{u \le t} V_u} - 1\right)$	Largest peak-to-trough capital loss [7].
Average Turnover	$\overline{\text{Turnover}} = \frac{1}{N}\sum_{t=1}^{N}\sum_{i=1}^{10}\lvert w_{i,t}^{\text{target}} - w_{i,t}^{\text{cur}}\rvert$	Mean daily weight reallocation across the 10 risky equities [5].
Total Cost Rate	$\text{Cost Rate} = \frac{\sum_{t=1}^{N} C_t}{V_0} \times 100\%$	Cumulative transaction fees paid, as a percentage of initial capital V_0.

Chapter 5 — Model Architecture and Mathematical Formulation

5.1 Actor–Critic Framework and Policy Choice

The portfolio allocation agents are developed within an Actor–Critic framework and optimized using Proximal Policy Optimisation (PPO) [4]. PPO is selected because: (i) it natively supports continuous stochastic distributions on the unit simplex manifold via the Dirichlet distribution [6]; (ii) the Dirichlet closed-form probability density enables exact evaluation of importance sampling ratios; (iii) the clipped surrogate objective prevents catastrophic policy shifts in noisy financial environments [4]; and (iv) it provides a modular architecture allowing controlled ablation of temporal encoders, reward functions, and regularization penalties.

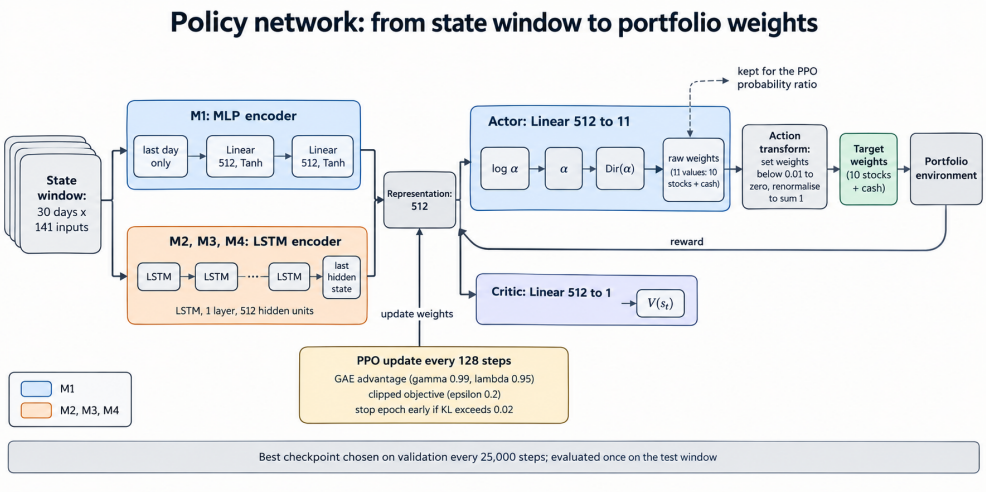


Figure 5.1: Actor–Critic architecture from state tensor to executed portfolio weights.

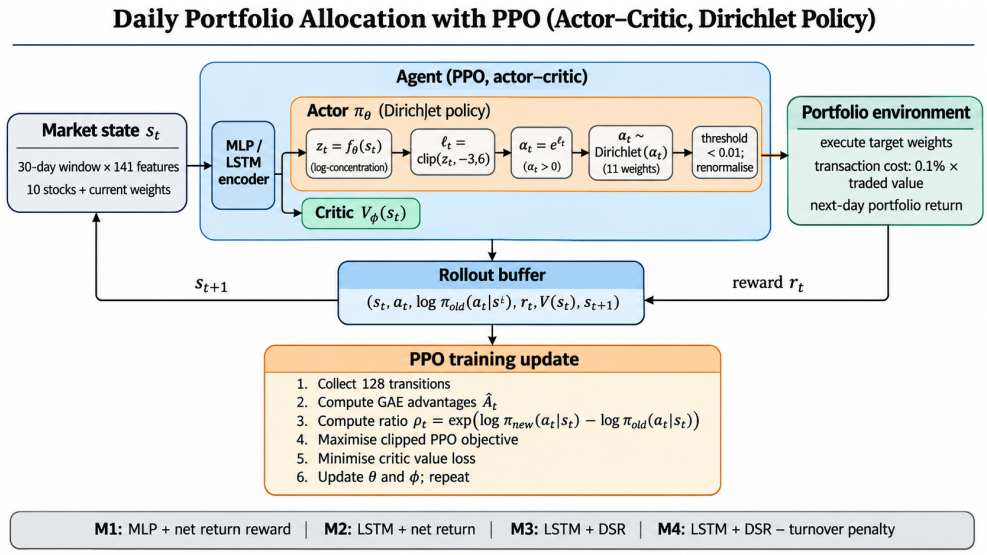


Figure 5.2: PPO Dirichlet allocation and environment interaction loop.

As shown in Figures 5.1 and 5.2, the actor and critic share an encoder backbone. The actor head outputs Dirichlet distribution parameters, while the critic head estimates the scalar state-value baseline $V_\phi(\mathbf{s}_t)$ for advantage computation.

5.2 Encoder Backbones: MLP versus LSTM

To evaluate the contribution of temporal sequence modeling, two encoder backbones are compared. Model M1 uses the instantaneous state $\mathbf{s}_t \in \mathbb{R}^{141}$ as input, whereas Models M2–M4 use the historical sequence $\mathbf{F}_t \in \mathbb{R}^{30\times 141}$ as input to the recurrent LSTM encoder:

- **MLP Encoder (Model M1):** A static two-layer feedforward network observing solely the instantaneous state $\mathbf{s}_t \in \mathbb{R}^{141}$. Intermediate latent representations $\mathbf{h}_{1,t}, \mathbf{h}_t \in \mathbb{R}^{512}$ and actor output $\mathbf{z}_t \in \mathbb{R}^{11}$ are computed

via:

$$\mathbf{h}_{1,t} = \tanh(\mathbf{W}_1 \mathbf{s}_t + \mathbf{b}_1), \tag{5.1}$$

$$\mathbf{h}_t = \tanh(\mathbf{W}_2 \mathbf{h}_{1,t} + \mathbf{b}_2), \tag{5.2}$$

$$\mathbf{z}_t = \mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a, \tag{5.3}$$

where $\mathbf{W}_1 \in \mathbb{R}^{512\times 141}$, $\mathbf{W}_2 \in \mathbb{R}^{512\times 512}$, and $\mathbf{W}_a \in \mathbb{R}^{11\times 512}$.

- **Recurrent LSTM Encoder (Models M2–M4):** A single-layer Long Short-Term Memory (LSTM) network with hidden size $H = 512$ processing the sliding sequence $\mathbf{F}_t = [\mathbf{s}_{t-T+1}, \ldots, \mathbf{s}_t] \in \mathbb{R}^{30\times 141}$ over time window $t - T + 1$ to t ($T = 30$ trading days) as input [7]:

$$(\mathbf{h}_t, \mathbf{c}_t) = \mathrm{LSTM}_\psi(\mathbf{F}_t), \quad \mathbf{z}_t = \mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a \quad [7], \tag{5.4}$$

  where $\mathbf{h}_t \in \mathbb{R}^{512}$ is the final hidden state summarizing the 30-day temporal trajectory from time step $t - T + 1$ to t.

- **Critic Value Head:** Sharing the identical 512-dimensional encoder latent representation $\mathbf{h}_t \in \mathbb{R}^{512}$ with the actor, the critic head parameterised by $\phi = \{\mathbf{W}_v, b_v\}$ maps the latent feature vector linearly to a single scalar state-value baseline estimate:

$$V_\phi(\mathbf{s}_t) = \mathbf{W}_v \mathbf{h}_t + b_v \quad \text{(for M1)}, \qquad V_\phi(\mathbf{F}_t) = \mathbf{W}_v \mathbf{h}_t + b_v \quad \text{(for M2–M4)}, \tag{5.5}$$

  where $\mathbf{W}_v \in \mathbb{R}^{1\times 512}$ is initialized orthogonally with gain 1.0, and scalar bias $b_v \in \mathbb{R}$ is initialized to zero (`nn.init.zeros_`), matching the source implementation (`self.critic = nn.Linear(hidden, 1)`). The critic produces the baseline value estimate V_ϕ used for advantage estimation in PPO.

5.3 Continuous Dirichlet Policy and Execution Logic

Target portfolio weights must satisfy the unit simplex constraints: $w_{i,t} \geq 0$ and $\sum_{i=1}^{11} w_{i,t} = 1$. Following Yang et al. [6], we parameterise the policy using a continuous multivariate Dirichlet distribution where positive concentration parameters $\boldsymbol{\alpha}_t \in \mathbb{R}^{11}_{>0}$ are formulated directly from the raw actor logits $\mathbf{z}_t \in \mathbb{R}^{11}$:

$$\boldsymbol{\alpha}_t = \exp(\mathbf{z}_t), \qquad \alpha_{i,t} = \exp(z_{i,t}), \quad i = 1, \ldots, 11. \tag{5.6}$$

During environment rollouts, raw continuous action vectors $\mathbf{a}_t \in \Delta^{10}$ are sampled via normalized Gamma variates: $g_{i,t} \sim \mathrm{Gamma}(\alpha_{i,t}, 1)$ with $a_{i,t} = g_{i,t} / \sum_j g_{j,t}$ [6]. During deterministic evaluation, the policy executes the analytical mean $\mathbb{E}[a_{i,t}] = \alpha_{i,t} / \sum_j \alpha_{j,t}$.

We explicitly distinguish between the raw action $\mathbf{a}_t$ sampled from the Dirichlet distribution and the executed portfolio weights $\mathbf{w}_t^{\text{target}}$ obtained after thresholding and renormalizing the raw action. To eliminate tiny fractional positions that incur wasteful transaction fees, an execution threshold $\tau = 0.01$ (1%) is applied: allocations below this threshold are set to zero, while the remaining allocations are renormalized to sum to one:

$$\widetilde{w}_{i,t} = \begin{cases} a_{i,t}, & a_{i,t} \geq \tau, \\ 0, & a_{i,t} < \tau, \end{cases} \qquad i = 1, \ldots, 11 \tag{5.7}$$

Then, the executed target weights are obtained via the normalization equation:

$$w_{i,t}^{\text{target}} = \frac{\widetilde{w}_{i,t}}{\sum_{j=1}^{11} \widetilde{w}_{j,t}}, \quad \text{where } \mathbf{a}_t \sim \mathrm{Dirichlet}(\boldsymbol{\alpha}_t) \quad [6]. \tag{5.8}$$

The direct probability density of the continuous multivariate Dirichlet distribution governing target allocations on the 11-dimensional portfolio simplex Δ^{10} is given by [6]:

$$\boxed{\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t) = \frac{\Gamma\left(\sum_{i=1}^{11} \alpha_{i,t}\right)}{\prod_{i=1}^{11} \Gamma(\alpha_{i,t})} \prod_{i=1}^{11} a_{i,t}^{\alpha_{i,t}-1}} \tag{5.9}$$

where $\mathbf{a}_t$ lies on the 11-dimensional unit portfolio simplex Δ^{10} and the concentration parameters satisfy $\alpha_{i,t} > 0$. Importantly, Equation (5.9) is evaluated strictly for the raw sampled action $\mathbf{a}_t$, rather than the thresholded and

renormalized executed weights $\mathbf{w}_t^{\text{target}}$. This preserves an exact, differentiable probability density for evaluating policy log-probabilities $\log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)$ and importance-sampling ratios $\rho_t(\theta)$ during PPO optimization, while the executed weights $\mathbf{w}_t^{\text{target}}$ are used by the environment for portfolio rebalancing. Here $\mathbf{s}_t$ denotes the state input to the policy (the instantaneous market observation $\mathbf{s}_t \in \mathbb{R}^{141}$ for the static feedforward baseline M1, or the sequence-encoded representation for recurrent LSTM models M2–M4 [7]).

Rebalancing from current weights $\mathbf{w}_t^{\text{cur}}$ to target weights $\mathbf{w}_t^{\text{target}}$ incurs linear transaction costs proportional to equity turnover ($c_{\text{trans}} = 0.001$, i.e., 0.1%): $\text{Turnover}_t = \sum_{i=1}^{10} |w_{i,t}^{\text{target}} - w_{i,t}^{\text{cur}}|$, with fee $C_t = c_{\text{trans}} V_t \cdot \text{Turnover}_t$ [7]. Portfolio wealth evolves after deducting transaction fees: $V_{t+1} = (V_t - C_t)G_t$, yielding realised net return $R_{\text{net},t} = (V_{t+1} - V_t)/V_t$. Weight drift across trading days is tracked causally using the gross portfolio return factor G_t:

$$G_t = \sum_{i=1}^{10} w_{i,t}^{\text{target}}(1 + R_{i,t+1}) + w_{\text{cash},t}^{\text{target}}, \quad w_{i,t+1}^{\text{cur}} = \frac{w_{i,t}^{\text{target}}(1 + R_{i,t+1})}{G_t}. \tag{5.10}$$

5.4 Reward Formulations across Models M1 to M4

The scalar reward r_t emitted to the agent defines our core ablation dimensions:

1. **Models M1 and M2 (Net Return):** $r_t = R_{\text{net},t}$. The agent directly maximises realised portfolio return after transaction costs.

2. **Model M3 (Differential Sharpe Ratio):** $r_t = D_t$, providing an online, recursive approximation of the Sharpe ratio [2]:

$$D_t = \frac{B_{t-1}\Delta A_t - \frac{1}{2}A_{t-1}\Delta B_t}{(B_{t-1} - A_{t-1}^2 + \epsilon_{\text{dsr}})^{3/2}} \quad [2], \tag{5.11}$$

   where running return moments A_t (first moment) and B_t (second moment) start at $A_0 = 0, B_0 = 0$ and update causally with decay $\eta_{\text{dsr}} = 0.01$ [2]:

$$\Delta A_t = R_{\text{net},t} - A_{t-1}, \qquad A_t = A_{t-1} + \eta_{\text{dsr}}\Delta A_t \quad [2], \tag{5.12}$$

$$\Delta B_t = R_{\text{net},t}^2 - B_{t-1}, \qquad B_t = B_{t-1} + \eta_{\text{dsr}}\Delta B_t \quad [2]. \tag{5.13}$$

   An epsilon safeguard $\epsilon_{\text{dsr}} = 10^{-8}$ is included in the denominator to prevent division by zero during early steps or flat periods.

3. **Model M4 (DSR + Quadratic Turnover Regularisation):** $r_t = D_t - \lambda_{\text{turn}} \sum_{i=1}^{10} (w_{i,t}^{\text{target}} - w_{i,t}^{\text{cur}})^2$, with $\lambda_{\text{turn}} = 0.1$ ($\lambda_{\text{turn}} = 0.1$ was selected through preliminary fine-tuning during training using validation performance) [5]. The quadratic penalty penalises large reallocations during policy training to suppress excessive churning.

5.5 PPO Objective and Policy Optimization

Policy parameters θ and critic parameters ϕ are optimized using Proximal Policy Optimisation [4]. PPO maximizes the clipped surrogate objective to update the actor policy:

$$\boxed{L^{\text{clip}}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(\rho_t(\theta)\hat{A}_t,\ \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right]} \quad [4], \tag{5.14}$$

where $\epsilon = 0.2$ is the clipping parameter [4, 7], and the probability ratio $\rho_t(\theta)$ is defined as:

$$\rho_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{s}_t)}, \tag{5.15}$$

conditioned on state $\mathbf{s}_t$ (instantaneous observation for M1 or the sequence-encoded representation for recurrent models M2–M4). The advantage $\hat{A}_t$ is computed using Generalized Advantage Estimation (GAE) [3]. The temporal-difference error at step t is:

$$\delta_t = r_t + \gamma V_\phi(\mathbf{s}_{t+1}) - V_\phi(\mathbf{s}_t), \tag{5.16}$$

and the advantage estimate over the rollout horizon is given by:

$$\boxed{\hat{A}_t^{\text{GAE}(\gamma,\lambda)} = \sum_{l=0}^{T_{\text{rollout}}-t-1} (\gamma\lambda)^l \delta_{t+l}} \tag{5.17}$$

where r_t is the reward at time t, $\gamma = 0.99$ is the discount factor, $\lambda = \lambda_{\text{GAE}} = 0.95$ is the GAE smoothing parameter, $V_\phi(\mathbf{s}_t)$ is the critic's value estimate, δ_t is the temporal-difference error, and $\hat{A}_t$ is the estimated advantage. In our implementation, the summation limit runs over the rollout buffer of length $T_{\text{rollout}} = 128$ ($l = 0, \ldots, T_{\text{rollout}} - t - 1$), and terminal-state handling cuts the recursion at segment boundaries by using the bootstrap value $V_\phi(\mathbf{s}_{\text{final}})$ and zeroing subsequent trace accumulation. PPO clipping limits the incentive for excessively large policy-ratio changes, helping stabilize policy optimization.

To jointly train the shared encoder, actor, and critic parameter sets, the network minimizes the combined multi-task loss function:

$$\mathcal{L}^{\text{total}}(\theta, \phi) = -L^{\text{clip}}(\theta) + c_1 L^{\text{value}}(\phi) - c_2 \mathcal{H}(\pi_\theta), \tag{5.18}$$

where $L^{\text{value}}(\phi) = \hat{\mathbb{E}}_t \left[(V_\phi(\mathbf{s}_t) - \hat{V}_t^{\text{target}})^2 \right]$ is the mean squared error loss of the state-value baseline critic with weight coefficient $c_1 = 0.5$, and $\mathcal{H}(\pi_\theta) = \hat{\mathbb{E}}_t \left[\mathcal{H}(\pi_\theta(\cdot \mid \mathbf{s}_t)) \right]$ is the analytical Dirichlet policy entropy bonus with coefficient $c_2 = 0.01$ encouraging distribution exploration. Both hyperparameter values ($c_1 = 0.5$ and $c_2 = 0.01$) are directly adopted from Table 1 of Zou et al. [7] in accordance with Schulman et al. [4].

We formally partition the network parameters into three disjoint parameter sets:

(i) **Shared Encoder Parameters (ψ):** Weights and biases of the shared representation backbone (`self.enc` in `src/models.py`, either the 2-layer MLP for M1 or the single-layer LSTM for M2–M4) producing latent embedding $\mathbf{h}_t \in \mathbb{R}^{512}$.

(ii) **Actor Head Parameters (θ):** Policy output layer weights and biases $\{\mathbf{W}_a, \mathbf{b}_a\}$ (`self.actor`) projecting $\mathbf{h}_t$ to Dirichlet concentration logits $\mathbf{z}_t \in \mathbb{R}^{11}$.

(iii) **Critic Head Parameters (ϕ):** Value output layer weights and biases $\{\mathbf{W}_v, b_v\}$ (`self.critic`) projecting $\mathbf{h}_t$ to scalar state-value estimate $V_t \in \mathbb{R}$.

Because the encoder is genuinely shared between actor and critic instances in our codebase, it receives back-propagated gradient contributions from both the actor surrogate objective and the critic value loss during optimization (`loss.backward()` in `src/ppo.py`). The shared encoder parameter update is:

$$\psi \leftarrow \psi - \eta \nabla_\psi \mathcal{L}^{\text{total}}, \tag{5.19}$$

with the exact gradient decomposition:

$$\nabla_\psi \mathcal{L}^{\text{total}} = -\nabla_\psi L^{\text{clip}} + c_1 \nabla_\psi L^{\text{value}} - c_2 \nabla_\psi \mathcal{H}. \tag{5.20}$$

The actor and critic head parameter updates are correspondingly:

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}^{\text{total}} = \theta - \eta \left(-\nabla_\theta L^{\text{clip}} - c_2 \nabla_\theta \mathcal{H} \right), \tag{5.21}$$

$$\phi \leftarrow \phi - \eta \nabla_\phi \mathcal{L}^{\text{total}} = \phi - \eta \left(c_1 \nabla_\phi L^{\text{value}} \right), \tag{5.22}$$

where $\nabla_\theta L^{\text{value}} = \mathbf{0}$ and $\nabla_\phi L^{\text{clip}} = \mathbf{0}$, ensuring that distinct heads are updated without double-counting shared parameters. In the implementation, a single unified Adam optimizer (`torch.optim.Adam(model.parameters(), lr=3e-4)`) applies these updates jointly across $\{\psi, \theta, \phi\}$ with global gradient norm clipping at 0.5 [7].

For an unclipped transition $J_t(\theta) = \rho_t(\theta)\hat{A}_t$, differentiating with respect to policy parameters θ yields the actor policy gradient:

$$\nabla_\theta J_t(\theta) = \hat{A}_t \nabla_\theta \rho_t(\theta) = \hat{A}_t \frac{\nabla_\theta \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{s}_t)} = \rho_t(\theta) \hat{A}_t \nabla_\theta \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t) \quad [4]. \tag{5.23}$$

Equation (5.23) demonstrates that the gradient step is directed by the advantage estimate $\hat{A}_t$ and scaled by the importance weight $\rho_t(\theta)$, adjusting the Dirichlet concentration mapping to shift probability density toward higher-yielding allocations on Δ^{10} [6], where the shared encoder processes the instantaneous state $\mathbf{s}_t$ for M1 or the historical sequence $\mathbf{F}_t$ for M2–M4. Optimization is carried out via Adam ($\eta = 3 \times 10^{-4}$) with gradient norm clipping at 0.5 [7] and early stopping when approximate KL divergence exceeds $1.5 \times \text{target_kl} = 0.03$ [4].

5.6 PPO Training Parameters

Table 5.1 summarizes the PPO hyperparameters and operational settings extracted directly from our codebase, along with the empirical or theoretical basis for each choice.

Table 5.1: PPO training parameters and operational settings (verified from source code).

Parameter	Description	Value	Selection Basis / Source
Discount factor (γ)	Return discount factor	0.99	Zou et al. (2024) [7] (Table 1)
GAE parameter (λ_{GAE})	Advantage trace decay	0.95	Schulman et al. (2017) [4]
Rollout horizon (T_{rollout})	Interaction steps per buffer	128	Zou et al. (2024) [7] (Table 1)
PPO clipping range (ϵ)	Policy ratio clipping threshold	0.2	Schulman et al. (2017) [4]; Zou et al. (2024) [7]
Value coefficient (c_1)	Critic loss weight	0.5	Zou et al. (2024) [7] (Table 1); Schulman et al. (2017)
Entropy coefficient (c_2)	Policy entropy bonus weight	0.01	Zou et al. (2024) [7] (Table 1); Schulman et al. (2017)
Optimiser	Parameter optimiser	Adam	Zou et al. (2024) [7]
Learning rate (η)	Adam learning rate	3×10^{-4}	Zou et al. (2024) [7] (Table 1)
Target KL (target_kl)	Approximate KL early-stop	0.02	Implementation safeguard [4]
Training epochs (K)	Update epochs per rollout	10	Schulman et al. (2017) [4]
Mini-batch size (B)	Batch size per SGD update	64	Schulman et al. (2017) [4]
Gradient clipping	Maximum gradient norm	0.5	Zou et al. (2024) [7] (Table 1)
Reward normalisation	Online reward scaling	Welford Std	Online Welford running standardisation
Total environment steps	Training budget per run	300,000	Target budget; executes 300,032 steps (2,344 × 128)
Validation interval	Checkpoint evaluation frequency	25,000	Validation selection interval
Transaction cost (c_{trans})	Proportional cost per turnover	0.001	Standard institutional rate [7] (0.1%)
Execution threshold (τ)	Action truncation threshold	0.01	Implementation setting (1% minimum position)
Window length (T), hidden (H)	Sequence length; LSTM units	30; 512	Zou et al. (2024) [7]
DSR rate (η_{dsr})	Moment adaptation rate	0.01	Online moment decay setting [2]
Turnover penalty (λ_{turn})	M4 regularisation weight	0.1	Validation tuning on Fold 1 [5]

Chapter 6 — Controlled Ablation Study Design

6.1 Specification of Ablation Models M1–M4

To evaluate the contribution of each algorithmic component without confounding factors, we adopt a single-factor ablation design. Four model configurations (M1 through M4) are specified such that each successive transition alters exactly one algorithmic mechanism while holding all other parameters, training budgets, random seeds, transaction costs, and data folds strictly invariant:

1. **Model M1 (MLP Baseline + Net Return):** Two-layer feedforward MLP observing solely instantaneous market observation $\mathbf{s}_t \in \mathbb{R}^{141}$ with net return reward ($r_t = R_{\text{net},t}$) and zero penalty ($\lambda_{\text{turn}} = 0$).
2. **Model M2 (LSTM Recurrent Encoder + Net Return):** Replaces the static MLP with a single-layer LSTM ($H = 512$) over sliding window $\mathbf{F}_t \in \mathbb{R}^{30\times 141}$. Reward remains net return ($r_t = R_{\text{net},t}$). Transition M1 → M2 isolates **temporal memory and sequence modeling** [7].
3. **Model M3 (LSTM Encoder + Differential Sharpe Ratio):** Retains the LSTM backbone, replacing net return with the recursive Differential Sharpe Ratio ($r_t = D_t$). Transition M2 → M3 isolates **risk-aware reward engineering** [2].
4. **Model M4 (LSTM + DSR + Quadratic Turnover Regularisation):** Retains the LSTM backbone and DSR reward, augmented by a quadratic weight-change penalty on risky assets ($r_t = D_t - \lambda_{\text{turn}} \sum_{j=1}^{10} (w_{j,t}^{\text{target}} - w_{j,t}^{\text{cur}})^2$) with $\lambda_{\text{turn}} = 0.1$. Transition M3 → M4 isolates **turnover regularisation** [5].

Table 6.1 summarises the systematic ablation matrix across all four configurations.

Table 6.1: Systematic ablation matrix across models M1 to M4.

Configuration Property	Model M1	Model M2	Model M3	Model M4
Encoder Backbone	MLP (2-layer)	LSTM (1-layer)	LSTM (1-layer)	LSTM (1-layer)
Temporal Window (T)	1 day (last row)	30 days	30 days	30 days
Latent Dimension (H)	512	512	512	512
Policy Head	Dirichlet Δ^{10}	Dirichlet Δ^{10}	Dirichlet Δ^{10}	Dirichlet Δ^{10}
Action Threshold (τ)	0.01	0.01	0.01	0.01
Reward Formulation	Net Return	Net Return	DSR (D_t)	DSR (D_t)
Turnover Regularisation	None ($\lambda = 0$)	None ($\lambda = 0$)	None ($\lambda = 0$)	Quadratic Penalty ($\lambda = 0.1$)
Environment Cost Model	0.1% per turnover	0.1% per turnover	0.1% per turnover	0.1% per turnover
Training Budget	300,000 steps	300,000 steps	300,000 steps	300,000 steps
Ablation Mechanism	*Baseline*	**+ Temporal**	**+ Risk-Aware**	**+ Turnover Penalty**

6.2 Control Invariants and Training Algorithm

Experimental controls are strictly enforced: (i) Actor and Critic heads are instantiated prior to the encoder to ensure identical initial head weights across models under the same seed; (ii) all models train for a nominal budget of 300,000 environment interaction steps, executing exactly 2,344 rollout iterations of horizon $T_{\text{rollout}} = 128$ for a total of 300,032 completed transitions, with identical Adam learning rate (3×10^{-4}), batch size (64), and 10 PPO epochs; (iii) all models experience identical market sequences and transaction costs; and (iv) programmatic assertions verify that each transition alters exactly one dimension. Algorithm 6.1 details the complete training workflow.

Algorithm 6.1 Continuous Dirichlet PPO Daily Portfolio Allocation

Require: Market indicators $\mathbf{X}$, returns $\mathbf{R}$, cost $c_{\text{trans}} = 0.001$, threshold $\tau = 0.01$, penalty λ_{turn}, DSR rate $\eta_{\text{dsr}} = 0.01$, PPO hyperparameters $(\gamma, \lambda_{\text{GAE}}, \epsilon, \eta = 3 \times 10^{-4})$.

Ensure: Optimised policy weights θ^* selected by peak validation Sharpe ratio.

Initialise shared Encoder ψ, Actor θ, Critic ϕ, Adam optimiser; $V_0 \leftarrow 10^6$, $\mathbf{w}_0^{\text{cur}} \leftarrow [0, \ldots, 0, 1]^\top$; DSR moments $A_0 \leftarrow 0, B_0 \leftarrow 0$; step counter $N_{\text{step}} \leftarrow 0, N_{\text{eval}} \leftarrow 25{,}000$, $\text{SR}_{\text{best}} \leftarrow -\infty$.
while $N_{\text{step}} < 300{,}000$ **do** ▷ Completes 2,344 rollouts $\times 128 = 300{,}032$ environment steps
  Reset rollout buffer $\mathcal{B} \leftarrow \emptyset$.
  for $t = 1$ **to** $T_{\text{rollout}} = 128$ **do**
    Observe window $\mathbf{F}_t \in \mathbb{R}^{30 \times 141}$; extract instantaneous state $\mathbf{s}_t \in \mathbb{R}^{141}$ (last row of $\mathbf{F}_t$).
    Compute latent state $\mathbf{h}_t \leftarrow \text{Encoder}_\psi(\mathbf{s}_t)$ for M1, or $\text{Encoder}_\psi(\mathbf{F}_t)$ for M2–M4.
    Compute raw logits $\mathbf{z}_t \leftarrow \mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a$ and concentration parameters $\boldsymbol{\alpha}_t \leftarrow \exp(\mathbf{z}_t)$.
    Sample raw continuous action $\mathbf{a}_t \sim \text{Dirichlet}(\boldsymbol{\alpha}_t)$ via Gamma variates; evaluate and record:
    $$\begin{aligned} \text{M1:} \quad & lp_t \leftarrow \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t), \quad V_t \leftarrow V_\phi(\mathbf{s}_t), \\ \text{M2–M4:} \quad & lp_t \leftarrow \log \pi_\theta(\mathbf{a}_t \mid \mathbf{F}_t), \quad V_t \leftarrow V_\phi(\mathbf{F}_t). \end{aligned}$$
    Apply execution threshold: $\widetilde{w}_{i,t} \leftarrow a_{i,t} \cdot \mathbf{1}_{a_{i,t} \geq \tau}$; set executed target weights $\mathbf{w}_t^{\text{target}} \leftarrow \widetilde{\mathbf{w}}_t / \sum_j \widetilde{w}_{j,t}$.
    Rebalance: $\text{Turnover}_t \leftarrow \sum_{i=1}^{10} |w_{i,t}^{\text{target}} - w_{i,t}^{\text{cur}}|$, fee $C_t \leftarrow c_{\text{trans}} V_t \text{Turnover}_t$.
    Advance environment: update wealth $V_{t+1} \leftarrow (V_t - C_t)\left(\sum_{i=1}^{10} w_{i,t}^{\text{target}}(1 + R_{i,t+1}) + w_{\text{cash},t}^{\text{target}}\right)$,
    net return $R_{\text{net},t} \leftarrow (V_{t+1} - V_t)/V_t$, and drifted weights $\mathbf{w}_{t+1}^{\text{cur}}$.
    Compute model reward: $r_t \leftarrow R_{\text{net},t}$ (M1, M2); $r_t \leftarrow D_t$ (M3); $r_t \leftarrow D_t - \lambda_{\text{turn}} \sum_{i=1}^{10} (w_{i,t}^{\text{target}} - w_{i,t}^{\text{cur}})^2$ (M4).
    Store $(\mathbf{F}_t, \mathbf{a}_t, lp_t, V_t, r_t)$ in $\mathcal{B}$ with raw action $\mathbf{a}_t$; $N_{\text{step}} \leftarrow N_{\text{step}} + 1$.
  end for
  Standardise rewards $r_t \leftarrow r_t / \sigma_{\text{Welford}}$; compute GAE advantages $\hat{A}_t$ with boundary cuts and returns $\hat{V}_t^{\text{target}}$.
  for epoch $= 1$ **to** 10 **do**
    for each mini-batch of size $B = 64$ from $\mathcal{B}$ **do**
      Evaluate ratio $\rho_i(\theta) \leftarrow \exp(\log \pi_\theta - lp_i)$ and approximate KL divergence; **break** if KL > 0.03.
      Update (ψ, θ, ϕ) via Adam ($\eta = 3 \times 10^{-4}$, grad clip 0.5), minimising $\mathcal{L}^{\text{total}}(\theta, \phi)$ (Eq. (5.18)).
    end for
  end for
  if $N_{\text{step}} \geq N_{\text{eval}}$ **or** $N_{\text{step}} \geq 300{,}000$ **then**
    Evaluate validation Sharpe SR_{val}; **if** $\text{SR}_{\text{val}} > \text{SR}_{\text{best}}$ **then** save checkpoint $\theta^* \leftarrow \theta$, $\text{SR}_{\text{best}} \leftarrow \text{SR}_{\text{val}}$ **end if**
    while $N_{\text{eval}} \leq N_{\text{step}}$ **do** $N_{\text{eval}} \leftarrow N_{\text{eval}} + 25{,}000$ **end while**
  end if
end while
Restore best checkpoint θ^* for final out-of-sample test evaluation.

Chapter 7 — Results and Discussion

7.1 Overall Out-of-Sample Performance

Out-of-sample evaluation was conducted across four chronological 2-year test windows spanning 2018 through 2025. Table 7.1 reports the pooled test performance metrics (averaged across four folds and three random seeds) for Models M1–M4 alongside the passive Equal-Weight (EW) benchmark.

Table 7.1: Pooled out-of-sample test performance across 4 folds and 3 seeds (mean ± standard deviation over seeds).

Model	Mean Two-Year Test Return (%)	CAGR (%)	Sharpe Ratio	Sortino	Max DD (%)	Cost (% of V_0)
M1 (MLP)	20.62 ± 5.14	9.90 ± 2.31	0.71 ± 0.13	1.03 ± 0.20	-19.50 ± 1.04	9.68 ± 0.10
M2 (LSTM)	28.23 ± 2.03	13.37 ± 0.88	0.92 ± 0.07	1.34 ± 0.09	-18.52 ± 1.35	4.27 ± 1.44
M3 (+ DSR)	26.93 ± 3.27	12.68 ± 1.56	0.92 ± 0.11	1.34 ± 0.17	-19.95 ± 0.81	4.14 ± 1.15
M4 (+ Turn. Pen.)	20.97 ± 4.19	10.09 ± 1.85	0.92 ± 0.17	1.37 ± 0.26	$\mathbf{-16.04 \pm 1.72}$	$\mathbf{0.97 \pm 0.15}$
Equal-Weight	**36.48**	**17.08**	**1.13**	**1.66**	−18.91	**0.61**

The primary empirical finding is that the passive Equal-Weight benchmark achieved the highest mean two-year test return (36.48%) and highest Sharpe ratio (1.13), outperforming all four reinforcement learning models. Among active RL agents, M2 and M3 achieved the highest returns (28.23% and 26.93%), while M4 achieved the lowest execution friction (0.97% cost) and shallowest maximum drawdown (−16.04%).

Figure 7.1 illustrates cumulative portfolio profit across the continuous 2018–2025 timeline (with starting capital V_0 = INR 1,000,000 reinvested sequentially), and Figure 7.2 portrays the corresponding peak-to-trough drawdowns.

7.2 Analysis of Ablation Steps

To assess whether observed differences are consistent across environments, paired differences $(b-a)$ were computed across all 12 (fold, seed) evaluation pairs. Table 7.2 reports the paired mean deltas and 95% bootstrap confidence intervals (obtained by resampling whole folds with replacement across 10,000 bootstrap draws).

Table 7.2: Paired ablation differences $(b - a)$ with 95% bootstrap confidence intervals across 12 (fold, seed) pairs.

Ablation Transition	Metric	Mean Delta	95% CI	Folds Improved	Pairs Improved
M1 → M2 (Temporal Memory)	Sharpe Ratio	+0.201	[+0.156, +0.246]	4/4	9/12
	Two-Year Return (%)	+7.61%	[+4.81%, +10.01%]	4/4	10/12
	Max Drawdown (%)	+0.98%	[+0.19%, +1.76%]	3/4	6/12
	Daily Turnover	−0.106	[−0.138, −0.070]	4/4	11/12
	Cost (% of V_0)	−5.41%	[−7.37%, −3.34%]	4/4	11/12
M2 → M3 (DSR Reward)	Sharpe Ratio	+0.003	[−0.295, +0.222]	3/4	7/12
	Two-Year Return (%)	−1.30%	[−9.33%, +4.45%]	2/4	6/12
	Max Drawdown (%)	−1.43%	[−5.36%, +0.81%]	3/4	6/12
	Daily Turnover	−0.002	[−0.019, +0.014]	2/4	7/12
M3 → M4 (Turnover Regularisation)	Sharpe Ratio	+0.005	[−0.236, +0.280]	2/4	6/12
	Two-Year Return (%)	−5.96%	[−13.29%, +3.37%]	1/4	3/12
	Max Drawdown (%)	**+3.90%**	[+1.45%, +7.00%]	**4/4**	**10/12**
	Daily Turnover	**−0.060**	[−0.082, −0.044]	**4/4**	**12/12**
	Cost (% of V_0)	**−3.17%**	[−4.01%, −2.43%]	**4/4**	**12/12**
M1 → M4 (All Combined)	Sharpe Ratio	+0.209	[+0.012, +0.386]	3/4	8/12
	Max Drawdown (%)	+3.45%	[+2.52%, +4.03%]	4/4	8/12
	Daily Turnover	−0.167	[−0.190, −0.144]	4/4	12/12
	Cost (% of V_0)	−8.71%	[−10.07%, −7.27%]	4/4	12/12

7.2.1 Step 1: Impact of Temporal Memory (M2 versus M1)

Transitioning from the MLP baseline (M1) to the LSTM recurrent encoder (M2) represents the clearest improvement in the ablation sequence. The pooled Sharpe ratio increases from 0.714 to 0.915 (+0.201, 95% CI [+0.156, +0.246]), improving across all four folds (4/4). Simultaneously, daily turnover decreases from 0.186 to

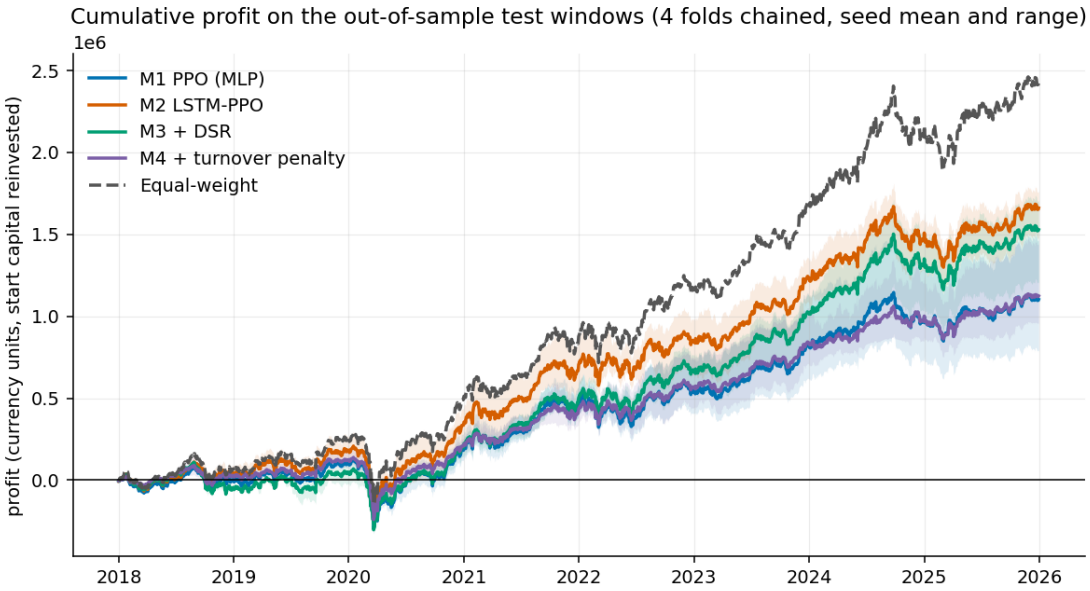


Figure 7.1: Cumulative out-of-sample profit across chained test windows (2018–2025, starting capital INR 1,000,000).

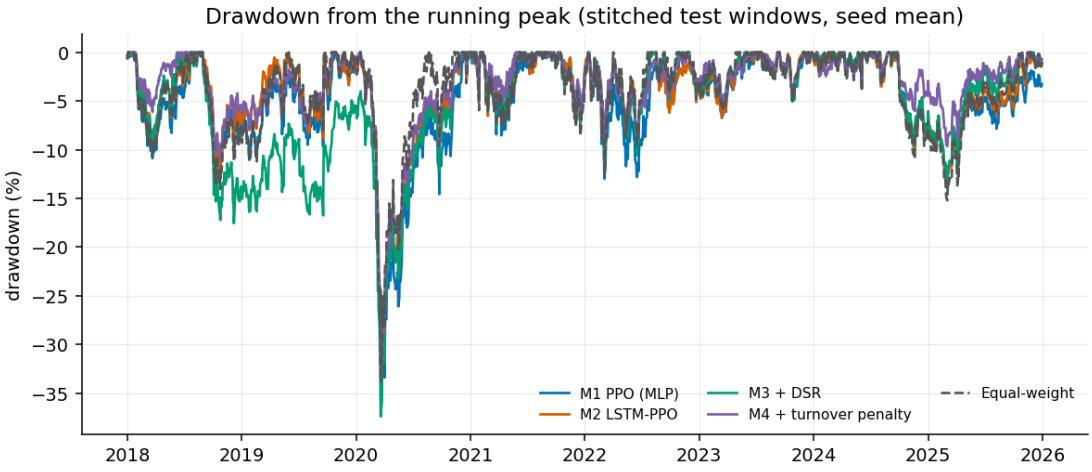


Figure 7.2: Out-of-sample portfolio drawdown from running peak across the chained timeline.

0.080, cutting cumulative transaction costs from 9.68% to 4.27% of capital. The 30-day LSTM representation provides valuable temporal smoothing relative to the single-day MLP baseline [7], tracking multi-period trends and reducing counterproductive churning.

7.2.2 Step 2: Impact of Differential Sharpe Ratio (M3 versus M2)

Replacing net return with the Differential Sharpe Ratio (M3) produces mixed, regime-dependent effects rather than a uniform improvement. The pooled Sharpe ratio remains flat (+0.003, 95% CI $[-0.295, +0.222]$), pooled return decreases from 28.23% to 26.93%, and maximum drawdown widens from −18.52% to −19.95%. Although M3 achieves strong Sharpe ratios in Folds 3 and 4 (1.328 and 1.164), its recursive variance penalty [2] dampens equity exposure during sustained upward trends, slightly curtailing profit capture.

7.2.3 Step 3: Impact of Turnover Regularisation (M4 versus M3)

Augmenting the reward with a quadratic penalty ($\lambda_{\text{turn}} = 0.1$) demonstrates an operational trade-off:

- **Friction Reduction:** Daily turnover decreases from 0.079 to 0.019 (a 76% reduction, improved in 12/12 pairs), reducing costs to 0.97% of starting capital, approaching the passive benchmark (0.61%).
- **Downside Protection:** Maximum drawdown improves by +3.90% (from −19.95% to −16.04%, 95% CI $[+1.45\%, +7.00\%]$), recording the shallowest drawdown among all models, including Equal-Weight (−18.91%).
- **Return Trade-off:** Conservatism reduces pooled return from 26.93% to 20.97% (improved in 1/4 folds). Regularisation limits churn and drawdown, but penalises rapid reallocations needed to capture explosive recoveries [5].

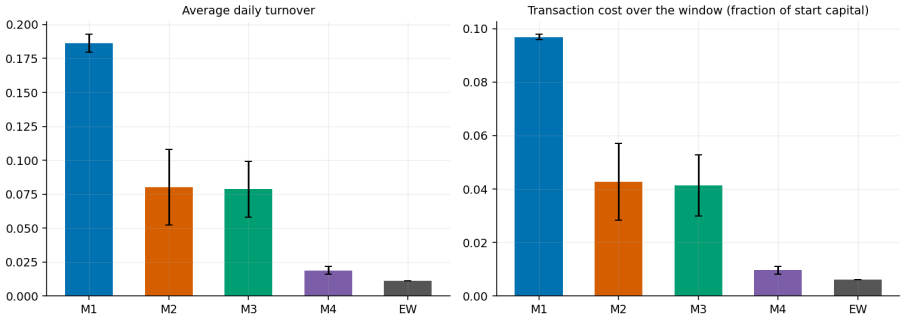


Figure 7.3: Average daily turnover and total transaction costs (% of initial capital V_0) across models.

7.3 Fold-by-Fold Performance Dynamics

Table 7.3 reports out-of-sample cumulative returns across individual two-year test windows, and Figure 7.4 illustrates cumulative profit trajectories over each fold.

Table 7.3: Out-of-sample cumulative return (%) per walk-forward fold (mean ± standard deviation over seeds).

Fold Window	Model M1	Model M2	Model M3	Model M4	Equal-Weight
Fold 1 (2018–2019)	10.0 ± 1.7	17.8 ± 7.7	4.4 ± 5.9	12.1 ± 5.3	**24.9**
Fold 2 (2020–2021)	34.0 ± 9.2	44.8 ± 6.9	44.1 ± 4.7	28.5 ± 14.8	**50.4**
Fold 3 (2022–2023)	23.4 ± 9.7	31.6 ± 4.7	34.5 ± 5.2	28.3 ± 5.3	**43.4**
Fold 4 (2024–2025)	15.0 ± 4.2	18.7 ± 1.3	24.7 ± 3.6	15.1 ± 4.0	**27.3**
Pooled Mean	20.6 ± 5.1	28.2 ± 2.0	26.9 ± 3.3	21.0 ± 4.2	**36.5**

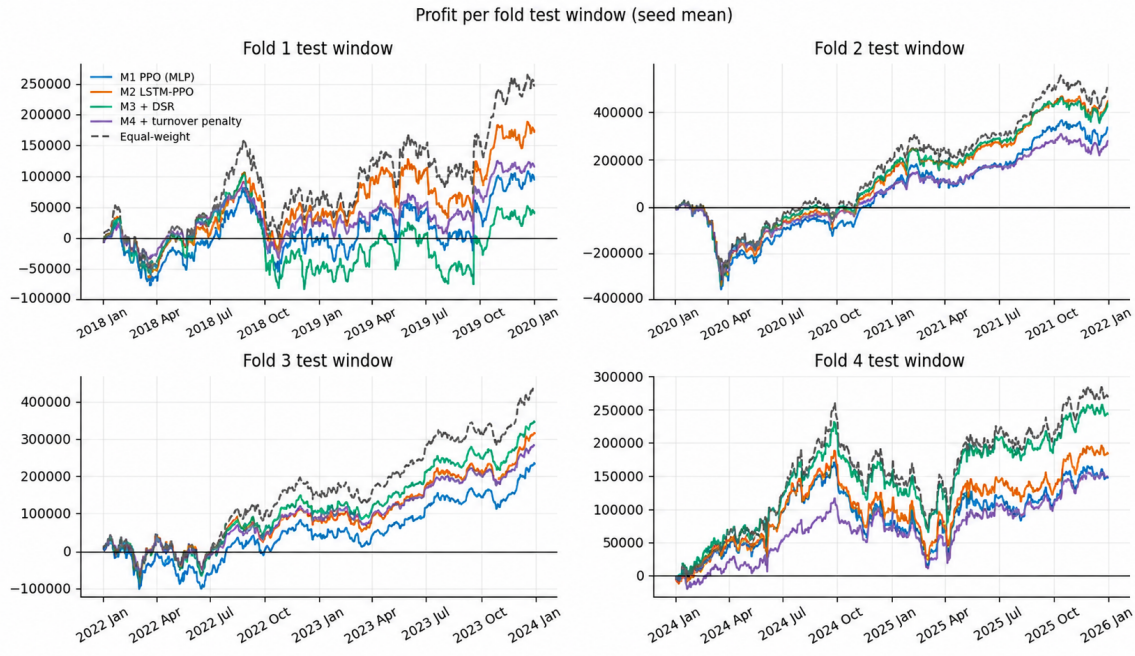


Figure 7.4: Cumulative out-of-sample portfolio profit across the four two-year test windows (mean across random seeds; initial capital V_0 = INR 1,000,000; y-axis denotes profit in INR; January 2026 marks the final tick for data ending December 2025).

Performance varies across market regimes:

- **Fold 1 (2018–2019):** Moderate growth; M2 leads RL agents (17.8%), M3 lags (4.4%), and Equal-Weight leads at 24.9%.
- **Fold 2 (2020–2021):** Pandemic shock and rapid recovery; M2 (44.8%) and M3 (44.1%) capture upside, whereas M4 lags (28.5%) due to constrained repositioning.
- **Fold 3 (2022–2023):** Tightening and sector rotation; M3 achieves its highest return (34.5%, Sharpe 1.328), outperforming M2 (31.6%) and M4 (28.3%).
- **Fold 4 (2024–2025):** Secular bull market; M3 delivers 24.7%, while M4 (15.1%) remains constrained.

7.4 Equal-Weight Benchmark Outperformance and Gap Decomposition

The passive Equal-Weight benchmark produced higher cumulative returns than every RL model across all test windows. We distinguish between the **mean two-year test return** (36.48% in Table 7.1) and the **compounded 8-year return across the chained 2018–2025 timeline** (135.66% in Table 7.4).

Table 7.4 decomposes the chained cumulative return gap between each model and Equal-Weight into two exploratory components: (i) **Cash Drag**: return forfeited by holding cash in an upward-trending market (evaluated against a counterfactual "EW Same Cash" portfolio); and (ii) **Stock Selection and Timing**: difference between model return and "EW Same Cash", reflecting active equity weights. Average asset allocations are presented in Table 7.5.

Table 7.4: Exploratory decomposition of cumulative out-of-sample return gap against Equal-Weight (% points over the chained 2018–2025 timeline).

Model	EW Return	Cash Drag	EW Same Cash	Stock Selection	Model Return
M1 (MLP)	135.66	−13.67	121.99	−2.11	119.88
M2 (LSTM)	135.66	−9.99	125.67	−2.14	123.53
M3 (+ DSR)	135.66	−15.14	120.52	−3.32	117.20
M4 (+ Turn. Pen.)	135.66	**−42.35**	93.31	−7.84	**85.47**

The decomposition indicates that cash drag accounts for the majority of the gap. The Indian equity market experienced strong overall growth over 2018–2025; the passive EW benchmark remained 100% invested at all times, capturing market beta with minimal turnover (0.61%). Conversely, Model M4 allocated an average of

Table 7.5: Average asset allocations across models over the out-of-sample evaluation period.

Model	RELI	TCS	HDFCB	ICICIB	ITC	HIND	LT	SUNP	MARU	BHAR	Cash
M1	0.088	0.120	0.085	0.087	0.084	0.077	0.091	0.087	0.087	0.094	**0.101**
M2	0.098	0.144	0.073	0.091	0.081	0.087	0.082	0.090	0.098	0.071	**0.087**
M3	0.089	0.119	0.073	0.077	0.072	0.079	0.073	0.094	0.107	0.086	**0.131**
M4	0.066	0.103	0.079	0.061	0.049	0.066	0.057	0.071	0.092	0.032	**0.324**
EW	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	**0.000**

32.4% to cash to avoid volatility and rebalancing penalties. Under the zero-return cash assumption, this resulted in an estimated −42.35% cash drag relative to EW, compared to −7.84% from equity selection. In addition, the selected large-caps represent corporate leaders that prospered over 2011–2025, creating a strong survivorship baseline that active strategies struggle to surpass after frictions.

7.5 M4 Hyperparameter Tuning and Validation Dynamics

Figure 7.5 depicts the parameter sweep over $\lambda_{\text{turn}} \in \{0.01, 0.03, 0.1, 0.3\}$ on Fold 1 validation data across two seeds (seeds 0 and 1).

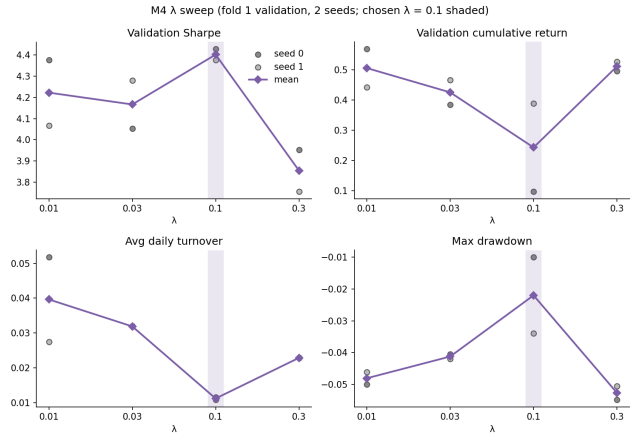


Figure 7.5: M4 turnover penalty parameter sweep on Fold 1 validation (chosen $\lambda = 0.1$ shaded).

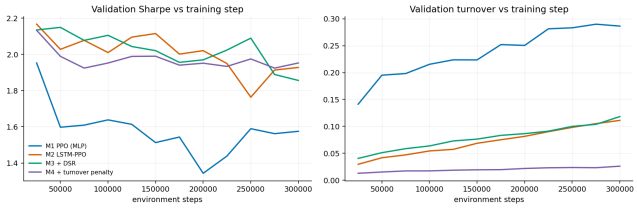


Figure 7.6: Validation Sharpe and turnover dynamics over 300,000 steps.

The preliminary sweep indicated that $\lambda = 0.1$ achieved a favorable trade-off on Fold 1 validation, reducing validation turnover while maintaining Sharpe ratio. At $\lambda = 0.01$, turnover remained elevated (0.040), whereas at $\lambda = 0.3$, reallocations were overly restricted. We disclose as an experimental limitation that $\lambda_{\text{turn}} = 0.1$ was selected based on preliminary validation tuning on Fold 1 alone across two seeds and applied frozen across all subsequent folds.

Finally, Figure 7.6 demonstrates that validation performance peaks between 50,000 and 100,000 steps before degrading, highlighting the necessity of checkpoint selection based on validation Sharpe ratio rather than relying on the final training step.

Chapter 8 — Conclusion

This study investigated the roles of temporal memory, risk-sensitive rewards, and turnover regularisation in deep reinforcement learning for daily portfolio allocation across ten major National Stock Exchange of India (NSE) equities over 2018–2025. Based on our four-fold, three-seed walk-forward experiments, we answer our core research questions:

- **RQ1 (Temporal Architecture):** *Does an LSTM recurrent encoder improve out-of-sample risk-adjusted returns and policy stability compared to an MLP baseline?*

  Answer: Yes. Under our setup, transitioning from the static MLP (M1) to the sequence-aware LSTM (M2) increased the pooled Sharpe ratio from 0.714 to 0.915 (+0.201, 95% CI [+0.156, +0.246]), showing gains across all four folds (4/4). The recurrent representation cut average daily turnover from 0.186 to 0.080 and reduced cumulative transaction costs from 9.68% to 4.27% of capital, confirming that multi-day sequence features filter daily noise and prevent erratic reallocations.

- **RQ2 (Risk-Aware Reward Shaping):** *Does transitioning from raw net return to the Differential Sharpe Ratio (DSR) improve Sharpe ratios and drawdowns?*

Answer: The effect is mixed and regime-dependent. The DSR reward (M3) produced a flat incremental Sharpe delta (+0.003, 95% CI $[-0.295, +0.222]$) relative to M2. While DSR performed strongly in trending regimes (Fold 3 Sharpe 1.328 and Fold 4 Sharpe 1.164), its recursive variance penalty slightly dampened mean two-year test return (26.93% vs. 28.23%) and did not improve pooled maximum drawdown (−19.95% vs. −18.52%).

- **RQ3 (Turnover Regularisation):** *Does augmenting the reward with a quadratic penalty suppress turnover and transaction costs, and what is the trade-off?*

  Answer: Yes, with an explicit operational trade-off. Incorporating the quadratic weight-change penalty in Model M4 reduced average daily turnover from 0.079 to 0.019 (a 76% reduction, improved in 12/12 pairs) and lowered transaction fees to 0.97% of capital. M4 also achieved the shallowest maximum drawdown among all evaluated models (−16.04%, +3.90% over M3). However, this conservatism reduced mean two-year test return from 26.93% to 20.97%, demonstrating that penalising turnover restricts participation in rapid market rebounds.

- **RQ4 (Benchmark Competitiveness):** *Can a continuous actor–critic DRL agent outperform a passively rebalanced equal-weight benchmark in a realistic Indian equity universe?*

  Answer: No. The passive Equal-Weight benchmark achieved superior mean two-year test return (36.48%) and Sharpe ratio (1.13). Our exploratory gap decomposition reveals that this is primarily driven by *cash drag* (−42.35% in M4) resulting from learned cash reserves in a secular bull market under the zero-return cash assumption, alongside survivorship bias among established large-cap leaders.

In conclusion, deep reinforcement learning can autonomously acquire disciplined, low-turnover allocation strategies on continuous simplices. While active outperformance against fully-invested broad-market beta remains difficult due to cash drag and trading costs, our controlled ablation isolates the empirical value of sequence modeling, reward shaping, and turnover control for quantitative asset management.

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